

Aditya Shrestha

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Research Interests

Machine Learning Systems: adaptive stream processing, backpressure control, and resource-aware ML infrastructure.

Education

Kathmandu University

B.E. in Computer Engineering

Dhulikhel, Nepal

Dec. 2022 – Expected Feb. 2027

- **Cumulative GPA:** 3.92 / 4.00 (through six semesters, Year I–III)
- **Semester GPA:** 3.78, 3.81, 3.95, 4.00, 4.00, 3.95 (two semesters at 4.00/4.00)
- **Coursework:** Neural Networks & Deep learning, Computer Vision, Artificial Intelligence, Digital Signal Processing, Operating Systems, Computer Architecture & Organization, Algorithms & Complexity, Database Management Systems, Communication & Networking, Principles of Programming Languages, Compiler Design, Software Engineering, Numerical Methods, Statistics and Probability, Discrete Structures

Kathmandu Model Higher Secondary School

Higher Secondary (Science)

Baghbazar, Kathmandu, Nepal

Completed 2022

- **CGPA:** 3.72 / 4.00

Honors and Awards

- **1st Place, Global IME AI/ML Hackathon (2026):** Built *BankSentinel*, an AI intrusion detection system for banking networks.
- **Academic Excellence Award (2022):** Kathmandu Model Higher Secondary School.

Certifications

- **AWS Certified Solutions Architect – Associate (SAA-C03)** Sept. 2026 – Sept. 2029
[View Credly Badge](#)
- **AWS Academy Graduate – Cloud Foundations**
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Research Experience

TopoRing: Topological Learning for Collusion Detection

Independent Research Project

Aug. 2026 – Present

Python, PyTorch Geometric, TopoNetX

- Benchmarked 8 graph and tabular architectures on 5.08M transaction records (IBM AMLworld) to classify candidate money-laundering cycles.
- Implemented simplicial-complex neural networks with algebraic lifters, enforcing boundary invariants for 2-cell message passing over arbitrary cycle topologies.
- Designed a leak-free nested cross-validation protocol over 18 protected account groups, evaluating 200 model checkpoints across 5 random seeds.
- Executed group-blocked permutation tests (10,000 permutations) with Holm correction and bootstrap 95% CIs to quantify evidence for higher-order topological biases.

KLStream: Backpressure-Adaptive Stream Processing

Independent Systems Research

2026 – Present

C++, Custom Runtime

- Built a from-scratch C++ stream processing runtime with lock-free SPSC/MPMC queues, cooperative scheduling, and bounded-memory execution.
- Developed an adaptive batching mechanism that dynamically resizes ML-inference batches using downstream queue occupancy (analogous to TCP AIMD congestion control).
- Validated a custom Isolation Forest implementation against scikit-learn (Spearman $\rho = 0.947$ on synthetic financial tick data).
- Evaluated across 30 paired runs per configuration, reducing P95 detection latency by $2.4\times$ under bursty loads at a measured accuracy cost.
- Extended backpressure control to adaptive Matryoshka-embedding truncation for vector search, identifying and resolving a critical baseline measurement bug.

Sentinel-GL: Multi-Sensor GLOF Precursor Detection

2026

Course-Affiliated Computer Vision Project

Python, PyTorch, Google Earth Engine

- Engineered a fail-closed multi-sensor satellite pipeline (Sentinel-1/2, MODIS, ERA5) across 20 Himalayan glacial lakes, eliminating legacy pseudoreplication via strict lake-level bootstrap resampling ($N = 5$).
- Benchmarked a Temporal-Spatial Masked Autoencoder (TS-MAE) against classical baselines under identical information budgets, establishing tree-based model dominance (Isolation Forest AUC 0.91 vs TS-MAE 0.68).
- Executed a pre-registered backtesting protocol on the catastrophic 2023 South Lhonak GLOF, transparently reporting a negative precursor detection result to set a benchmark for reproducible remote sensing.

Selected Projects

BankSentinel: Multi-Agent Intrusion Detection

June 2026

Global IME AI/ML Hackathon (1st Place)

Python, BiLSTM, Isolation Forest

- Designed a five-agent pipeline combining Random Forests, regime-aware Isolation Forests, and a BiLSTM autoencoder for zero-day intrusion detection.
- Reduced false positives from 22.7% to 2.1% and cut alert volume by 87% via a four-layer suppression engine (validated on CICIDS-2017 and CTU-13).

CoreMesh: NUMA-Aware Scheduler for ARM64

Jan. 2026

Group Systems Project

C++17, ARM64

- Designed a consistent-hashing task scheduler with lock-free ring buffers and four-tier NUMA-aware work-stealing for many-core ARM64 systems.
- Benchmarked against Linux CFS across 13 configurations: achieved $1.37\times$ peak speedup and 100% completion vs. CFS's 92.3% at the 2M-task stress scale.

Flood Sentinel: Self-Supervised Flood Forecasting

2026

Independent Research

Python, PyTorch, TCN

- Built a 34-year multi-sensor benchmark across 54 CONUS basins, integrating USGS hydrometry, gridMET meteorology, and SNODAS snowpack with strict latency constraints to prevent lookahead bias.
- Designed a hybrid Causal TCN-Transformer (C-ENCODER, 147K params) that matches fully-supervised EA-LSTMs (AUC 0.83) and drastically outperforms them under extreme label scarcity (10% budget).
- Falsified the hypothesis that temporal anomaly scores alone provide actionable early warning lead times without hydrodynamic routing, documenting a crucial scientific negative finding.

FlowGate: Load-Adaptive IIR Filtering

July 2026

Course-Affiliated DSP Project

Python, Signal Processing

- Designed a load-adaptive IIR filter driven by an exogenous system-load signal, deriving closed-form Z-domain stability proofs and group-delay formulas.
- Evaluated on real crypto tick data (150 seeds), quantifying a load-shedding mechanism's 76.1% throughput gain at a $\sim 10\%$ ROC-AUC cost.

Technical Skills

- Languages:** C++ (17/20), Python, C, JavaScript/TypeScript, SQL
- Cloud & Infrastructure:** AWS (VPC, EC2, S3, RDS, Lambda, IAM), Docker, Linux
- Systems:** Lock-free data structures, stream processing runtimes, NUMA-aware scheduling, ARM64 architecture
- ML/Data:** PyTorch, PyTorch Geometric, scikit-learn, Pandas, Isolation Forest, BiLSTM
- Tools:** Git, CMake, PostgreSQL, FastAPI, React

Languages

Nepali (Native) , English (Fluent) , Hindi (Fluent) , Newari (Conversational)